Discussion Worksheet for Lectures 4 and 5

September 14, 2025

1 Key Points Summary

1.1 Lecture 4: Taylor's Theorem and Error Analysis

- **Taylor's Theorem**: For a function $f \in C^{n+1}[a,b]$, $f(x) = P_n(x) + R_n(x)$, where $P_n(x)$ is the n-degree Taylor polynomial around x_0: $f(x_0) + f'(x_0)(x-x_0) + \cdots + \frac{1}{n!}f^{(n)}(x_0)(x-x_0)^n$.
- **Remainder** $R_n(x)$: $\frac{1}{(n+1)!}f^{(n+1)}(\xi)(x-x_0)^{n+1}$ for some ξ between x and x_0. For integrals, bound R_n using $\max |f^{(n+1)}|$.
- **Examples**: For $\cos x$ at $x_0 = 0$, $P_2(x) = 1 - \frac{1}{2}x^2$, $R_2(x) = \frac{1}{3!}\cos(\xi)x^3$. For $\int e^{-t^2}dt$, do Taylor expansion of e^x and then use substitution $y = -x^2$ to estimate error better.

1.2 Lecture 5: Round-off Errors and Computer Arithmetic

- **Floating Point Representation**: Numbers as $(-1)^s \cdot 2^c \cdot (1+f)$, IEEE double: 1 sign, 11 exponent, 52 fraction bits. Machine epsilon $\varepsilon = 2^{-52} \approx 2.22 \times 10^{-16}$ (16 digits).
- **Rounding**: To even in ties (last digit 0). Underflow ($< 2^{-1022} \to 0$), overflow ($> 2^{1023} \to \infty$).
- **Errors**: Absolute $|p - p^*|$, relative $|p - p^*|/|p|$. Significant digits t if relative error $\leq 5 \times 10^{-t}$.
- **Cancellation**: In subtraction of close numbers, loss of digits. Example: Quadratic roots with large b in $x^2 - bx + c = 0$.
- **Sources**: Discretization (continuous to discrete) and round-off (finite precision). Multiplying large numbers causes overflow.

2 Exercises

1. **Taylor Expansion and Error (Lecture 4)**: Expand $f(x) = e^x$ around $x_0 = 0$ to degree 2. Compute $P_2(0.1)$ and bound $|R_2(0.1)|$ using $|f^{(3)}(\xi)| \leq e^{0.1} \approx 1.11$.

2. **Integral Approximation (Lecture 4)**: Approximate $\int_0^{0.2} \cos(t)dt$ using $P_2(t)$ for $\cos t$ around 0. Estimate the error using R_2.

3. **Floating Point and Relative Error (Lecture 5)**: In a system with machine epsilon $\varepsilon = 10^{-5}$ (5 digits), approximate $\pi \approx 3.14159$. Compute relative error for $p^* = 3.1416$ and determine significant digits.

4. **Rounding Example (Lecture 5)**: In a 6-bit system (1 sign, 3 exponent, 2 fraction), add 1.25×2^{-1} (binary 0.101) and 0.75×2^{-1} (binary 0.011). Show steps and apply round to even if tied.

5. **Cancellation in Quadratic (Lecture 5)**: Solve $x^2 - 200x + 1 = 0$ using standard formula (4 digits precision). Show cancellation for small root and recompute with rationalized form. Compute relative errors.